

Transport Ticketing Optimization

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Abstract

Public transportation systems worldwide are exploring innovative ticketing solutions to enhance service efficiency and passenger satisfaction. In this study, I investigate two novel strategies aimed at optimizing ticket validity and pricing: transport tickets with two distinct validity time slots and the implementation of dynamic pricing based on demand and travel time. Utilizing passenger flow data in Vilnius, I analyze the impact of these strategies on passenger costs and system revenue. The findings suggest that tickets with flexible validity periods can better accommodate diverse passenger needs, leading to more balanced utilization of public transport resources. Additionally, dynamic pricing appears effective in managing demand, reducing congestion during peak hours, and optimizing revenue. This study highlights the potential of these approaches to improve public transportation systems' efficiency and responsiveness to passengers' needs.

Motivation and goal

By basing ticket prices on the time spent in transit, fares can more accurately reflect the value of the service provided. Passengers traveling shorter distances or durations do not subsidize those on longer journeys, making the system fairer. This method can be especially appealing in urban areas where distances between stops can vary significantly.

Hypothesis

Validity period of a transport ticket can extend beyond a single, continuous time frame to include 2 distinct time slots, potentially with varying conditions for route changes. This flexibility could better accommodate passengers' diverse travel needs and preferences, optimizing utility and satisfaction.

Math Problem

Given a passenger flow data, determine the optimal time divider n (in minutes) that splits the day into two distinct time slots. Consider factors such as average travel duration and travel patterns. The objectives are following:

- **Optimal Divider Identification:** Identify the optimal time divider n (in minutes) that results in an approximately equal distribution of passengers purchasing short-duration tickets (valid for travel before n minutes) and long-duration tickets (valid for travel after n minutes)
- **Current Disbalance Assessment:** Assess whether there is a disbalance in the current system where the time divider is set at $n=30$ minutes. Determine if this setting leads to an unequal distribution of passengers purchasing short-duration versus long-duration tickets.

Dataset_GTFS_data

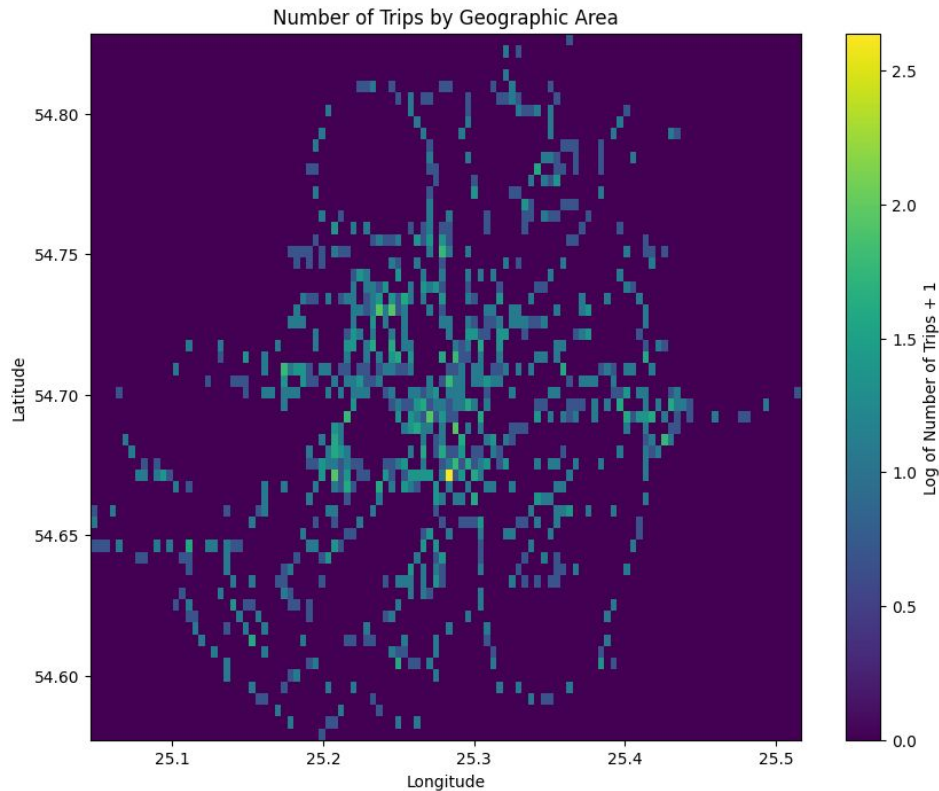
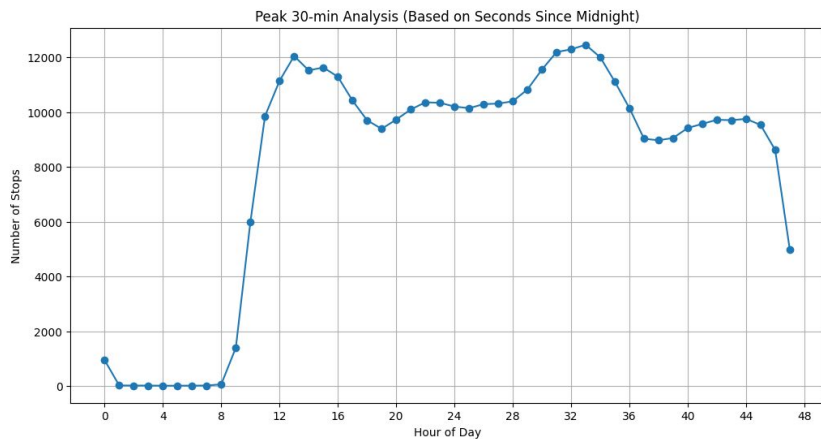
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...	...	...	...	...	...	...	...	...	...	...	...
100	vilnius_trol_17	vilnius	17	Žirmūnai-Naujininkai	NaN	800	https://www.stops.lt/vilnius/index.html#vilniu...	DC3131	FFFFFF	3001700	17 Žirmūnai-Naujininkai
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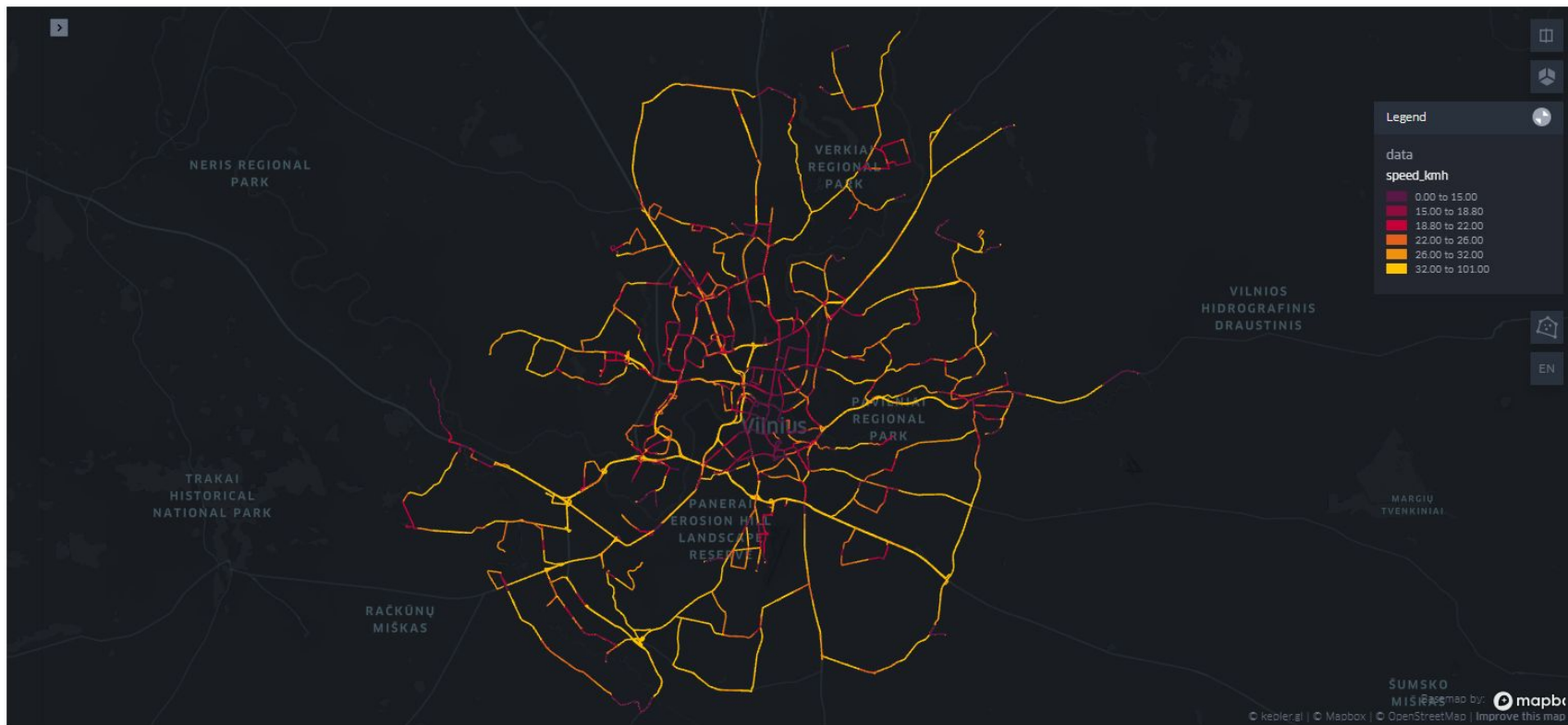
It also includes information about trips, stops, stop times, stops frequency, line frequency, average segments, etc.

Dataset_traffic

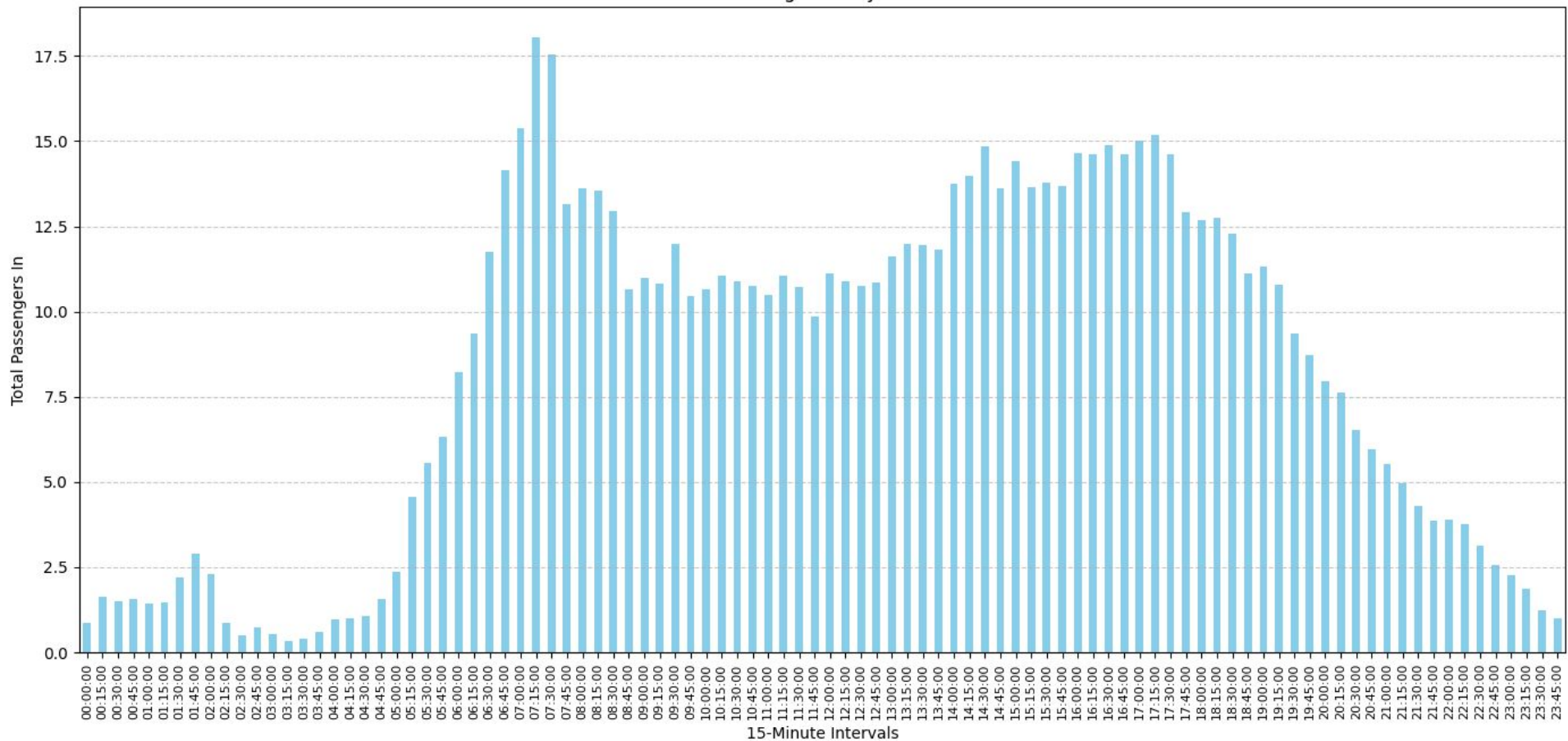
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4	2018-01-01 11:06:09	ANTAKALNIS-PARKO G.	18	2353	Respublikinė psichiatrijos ligoninė	0	0	7.00	54.69	25.42
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8	2018-01-01 19:17:52	STOTIS-N.VILNIA	31	2353	Respublikinė psichiatrijos ligoninė	0	0	4.00	54.69	25.42
9	2018-01-01 19:30:49	ANTAKALNIS-PARKO G.	18	2353	Respublikinė psichiatrijos ligoninė	0	0	2.33	54.69	25.42
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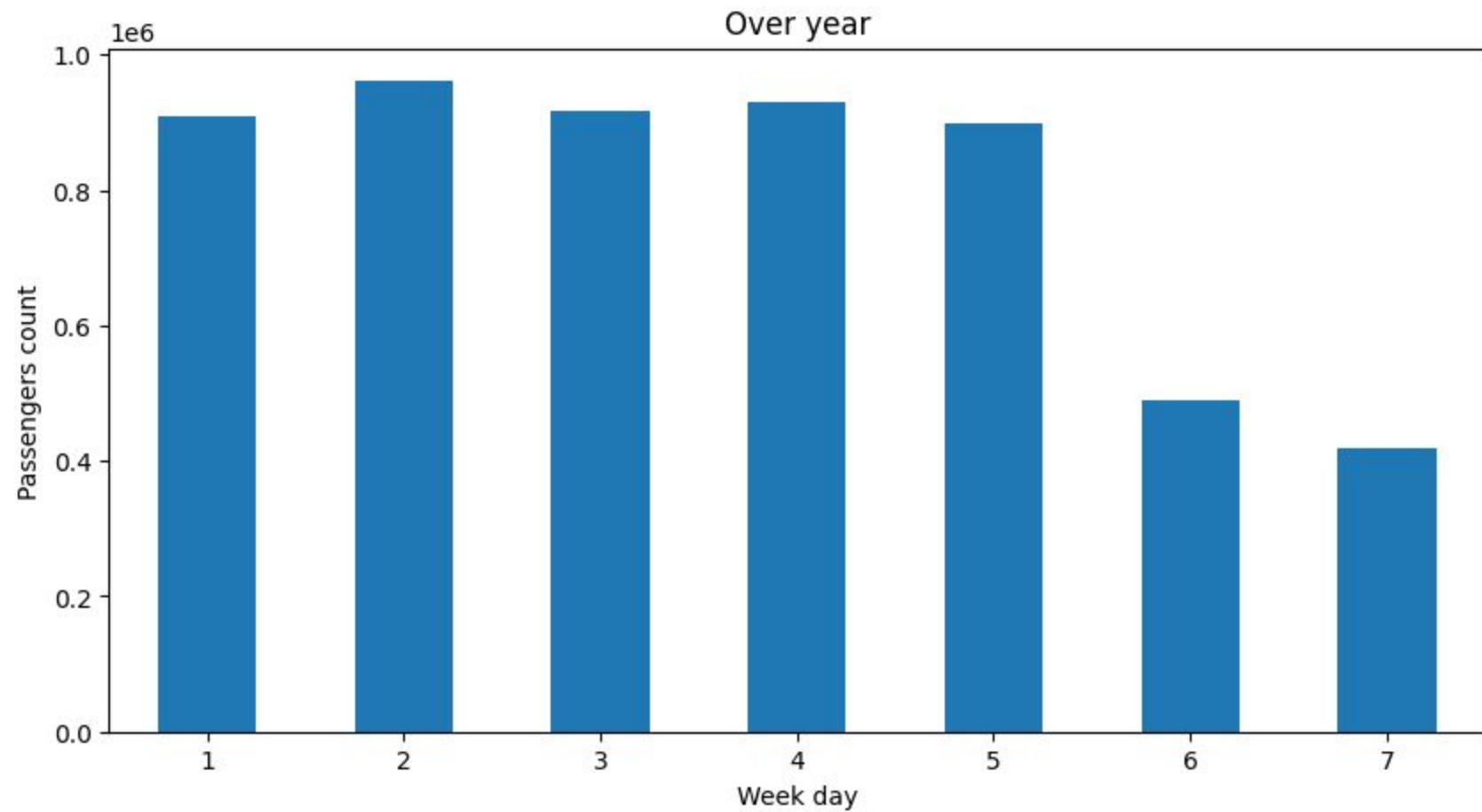
Analysis based on gtfs data

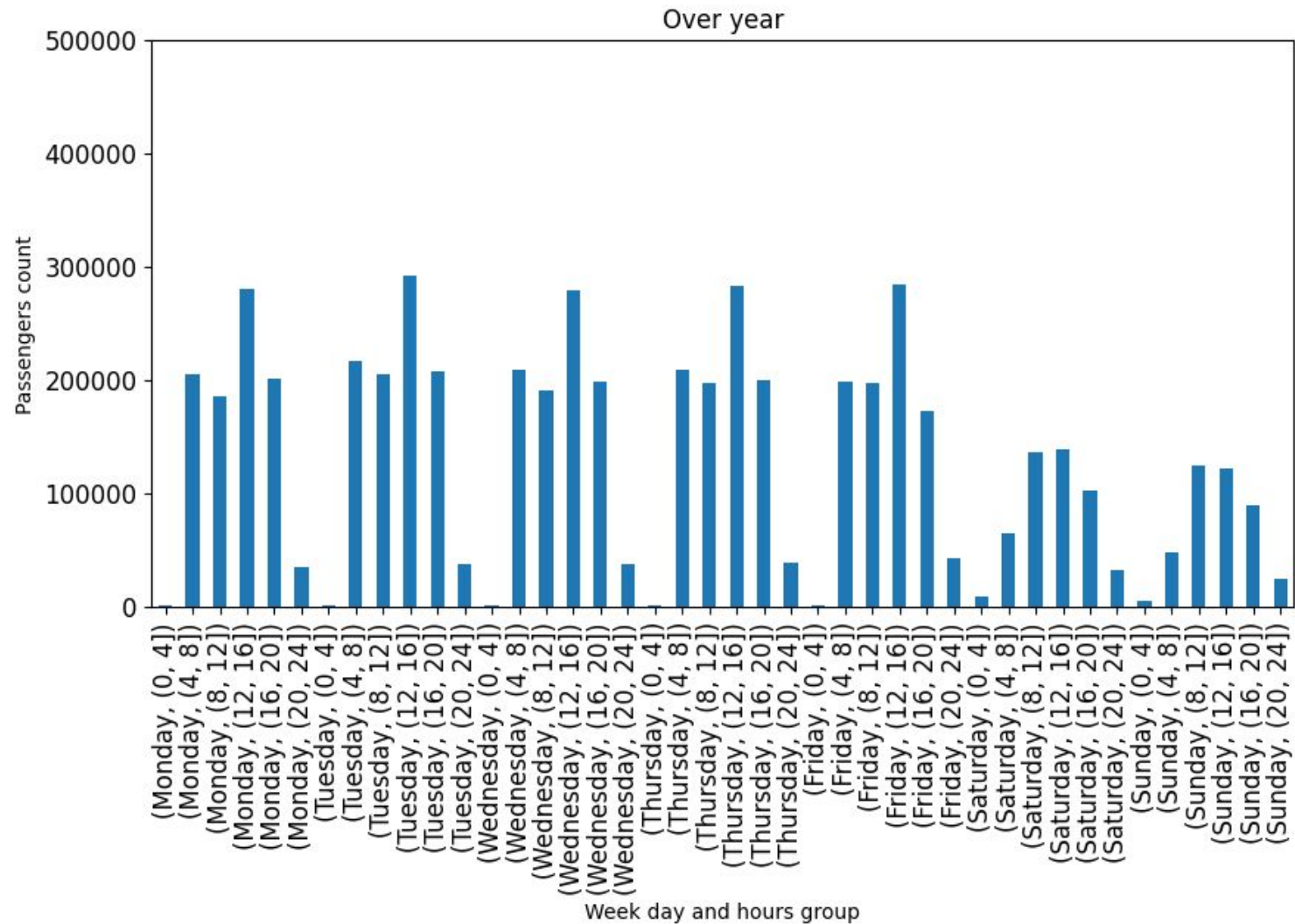




Distribution of Passengers In by 15-Minute Intervals







Analysis based on passenger traffic data

[Link to the monthly/hourly passengers flow heatmaps](#)

Solution Approach

Identify High-Traffic Stops:

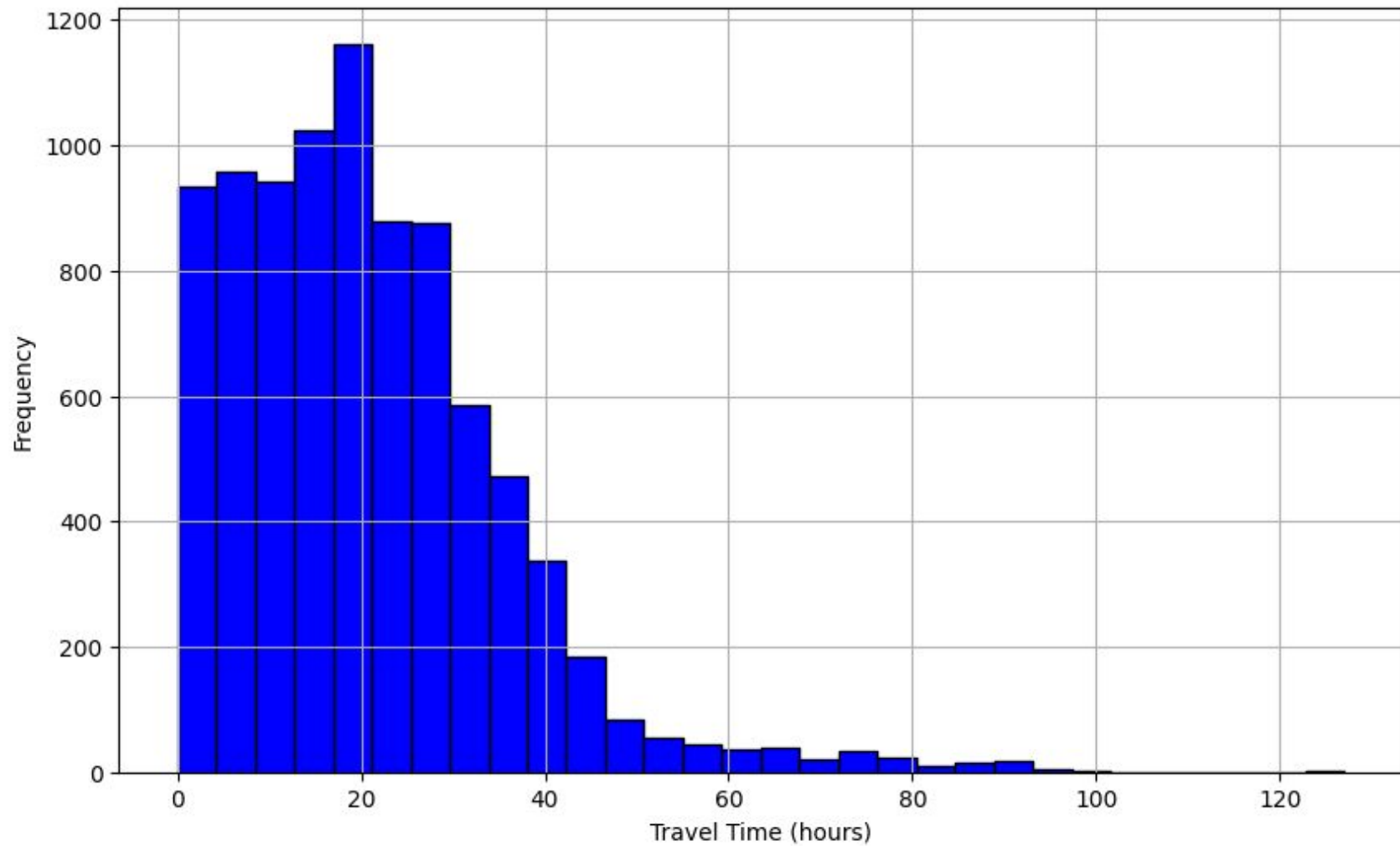
Analyze passenger data to determine the most frequently used stops, focusing on those that are utilized more than 25 times per day on average, separately for boarding (inbound) and alighting (outbound) passengers.

Solution Approach

Calculate Route Travel Times:

For each pair of high-traffic stops $stop_i$ and $stop_j$, calculate the travel time using the Google Maps API. This should be done for all possible routes between these stops.

Distribution of Travel Times

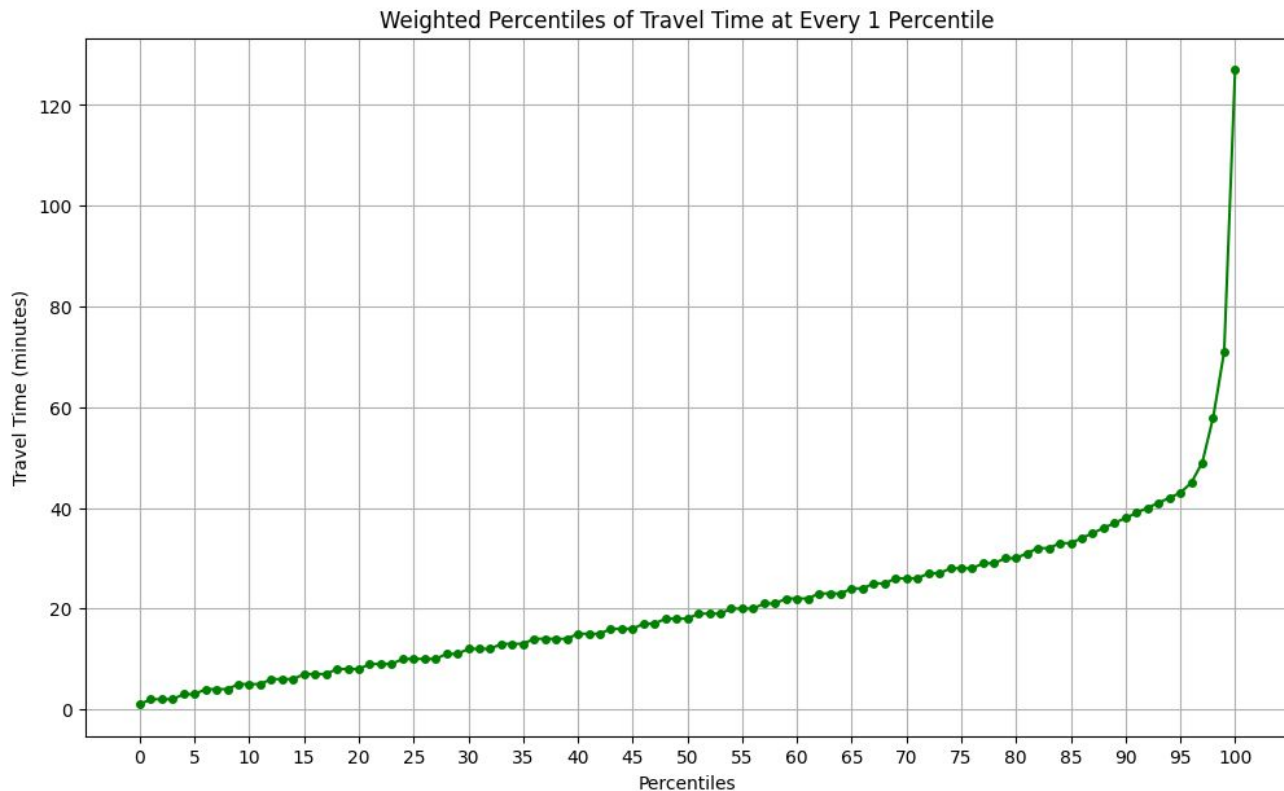


Solution Approach

Statistical Analysis of Travel Times:

- Compute the weighted mean travel time for each route, taking into account the frequency of trips or passenger volume.
- Additionally, calculate the median travel time and percentiles from 1st to 100th to provide a comprehensive statistical summary of travel times across all routes.

Results



Mean: 20.38

Median: 18.0

Hypothesis 2

Implementing dynamic pricing for public transport tickets, where fares fluctuate based on real-time demand and specific travel times, could lead to more efficient distribution of passengers across different time slots. This approach may reduce peak-time congestion, optimize revenue, and increase overall passenger satisfaction by providing more cost-effective options during off-peak hours.

Math Problem 2

Create a dynamic pricing formula that adjusts the cost of a transport ticket based on real-time demand and travel time. The formula should include variables for base fare, demand multiplier (dependent on current demand relative to baseline), and travel time multiplier (accounting for duration over a threshold). Solve for the fare during different scenarios, such as peak hours with high demand and off-peak hours with low demand.

Solution Approach

Formula for dynamic pricing:

$$p = p_0 * f(d) * g(tt),$$

Where:

p_0 – base fare

$f(d)$ – demand coefficient (function of demand with mean 1)

$g(tt)$ – time travel coefficient (function of travel time with mean 1)

Solution Approach

- The confidence interval for the distribution of prices should not vary by more than a factor of two. This ensures that the pricing remains fair and predictable across different demand levels and travel times.
- The mean price across the distribution should remain consistent with the base fare p_0 , ensuring that the overall pricing structure does not lead to an unintended increase or decrease in revenue.

Desired Properties for $f(d)$ and $g(tt)$

Symmetry Around the Median:

The functions $f(d)$ and $g(tt)$ should be symmetric around the median. This means that the deviation from the median should be identical in magnitude on both sides, ensuring balanced price adjustments.

Normal Distribution-like Behavior

Symmetry Around the Median:

The deviation of the functions from the median should resemble the shape of a normal distribution. The functions should increase or decrease smoothly as t moves away from the median, with values near the 50th percentile being more common and extreme percentile values (closer to 0 or 100) being less frequent.

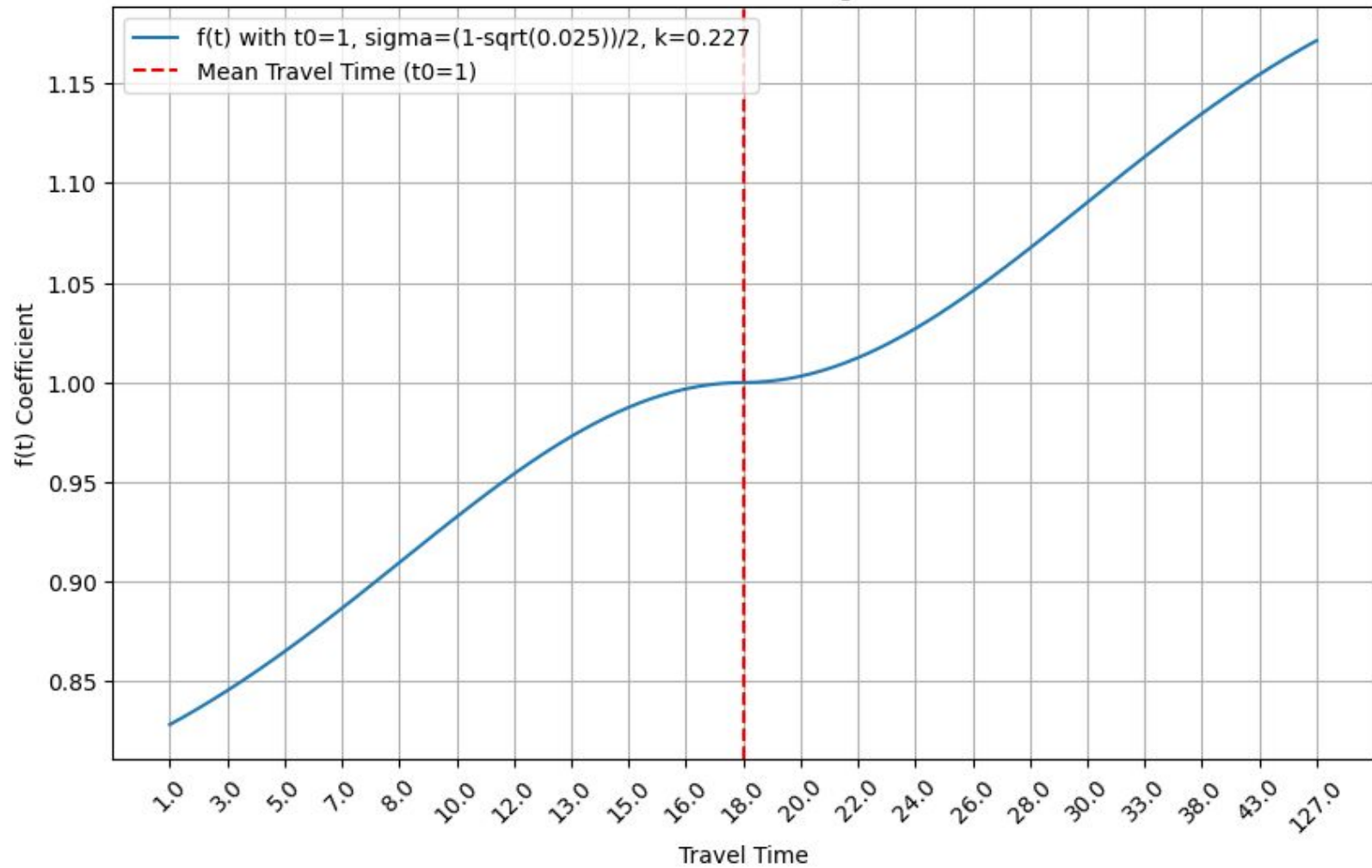
Finalized Function

$$f(t) = 1 + K * \text{sign}(t - t_0) * \left(1 - e^{-\left(\frac{|t-t_0|}{\mu}\right)^2}\right)$$

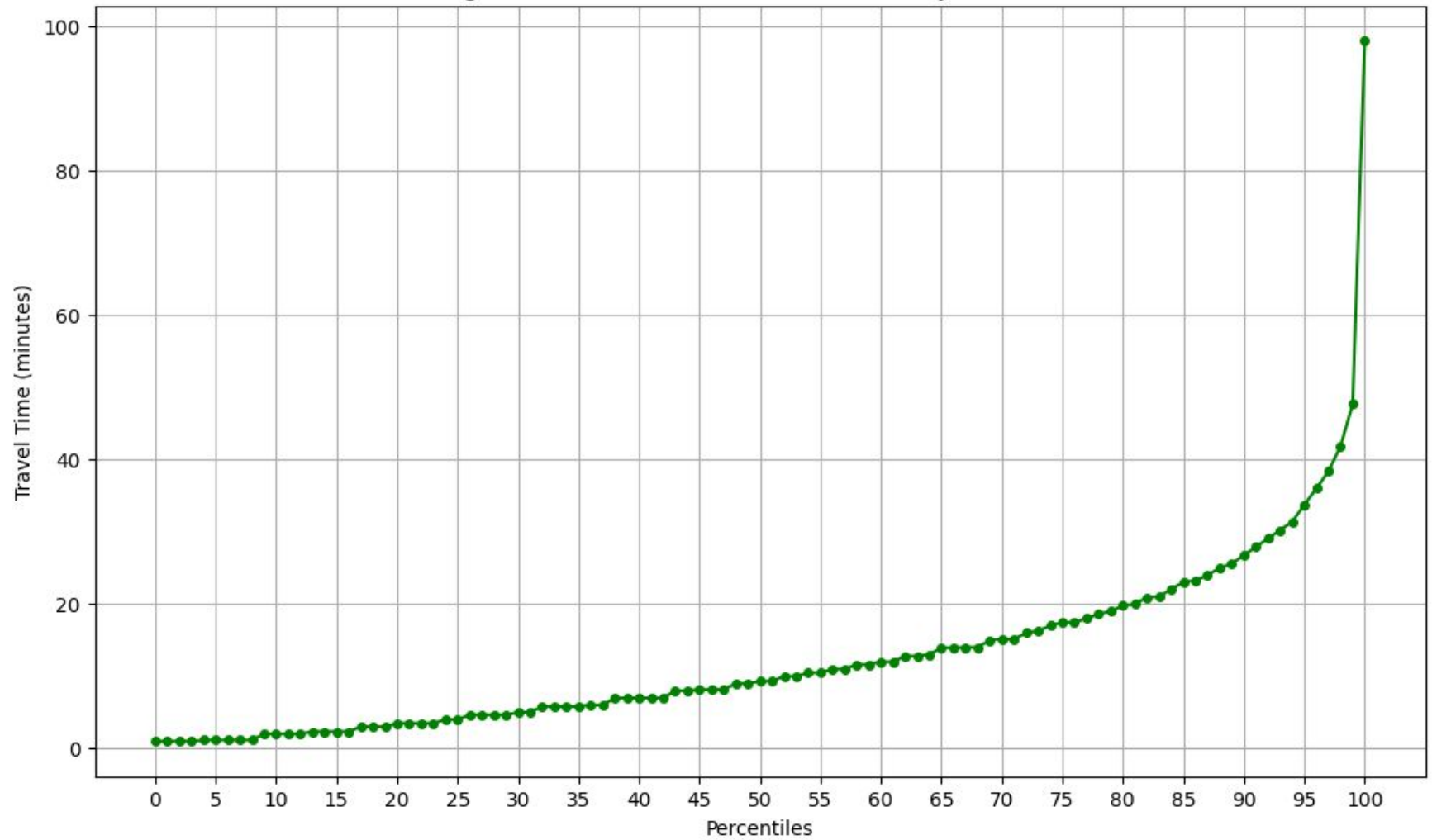
where:

- t is the scaled $(/100 + 0.5)$ percentile value for travel time (similarly: d for demand)
- $t_0 = 1$ represents the median
- K is a scaling factor to control the intensity of the price adjustment.
- $\mu = (1 - \sqrt{0.025})/2$ after confidence interval calculation

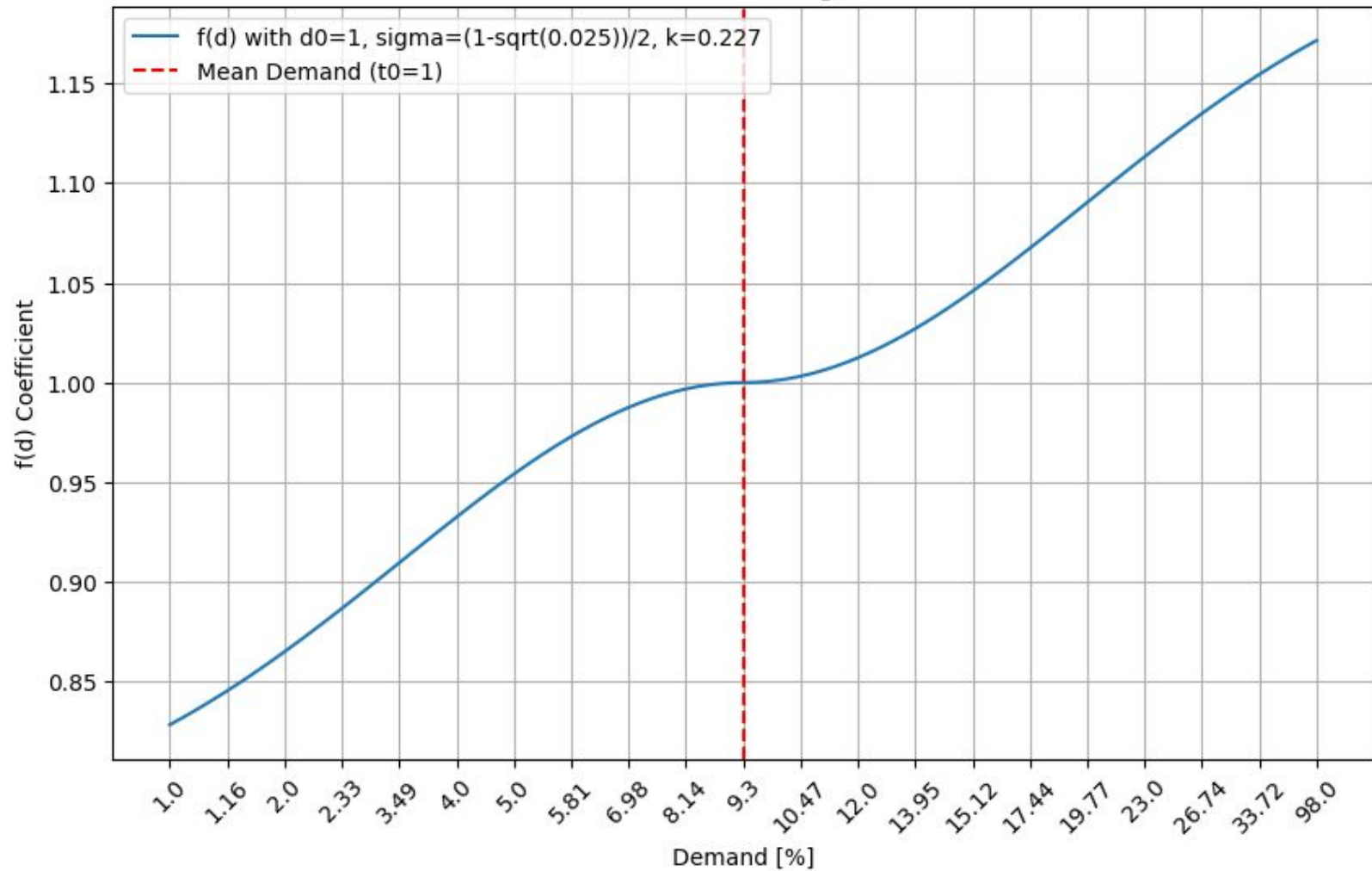
Travel-Time Based Pricing Function



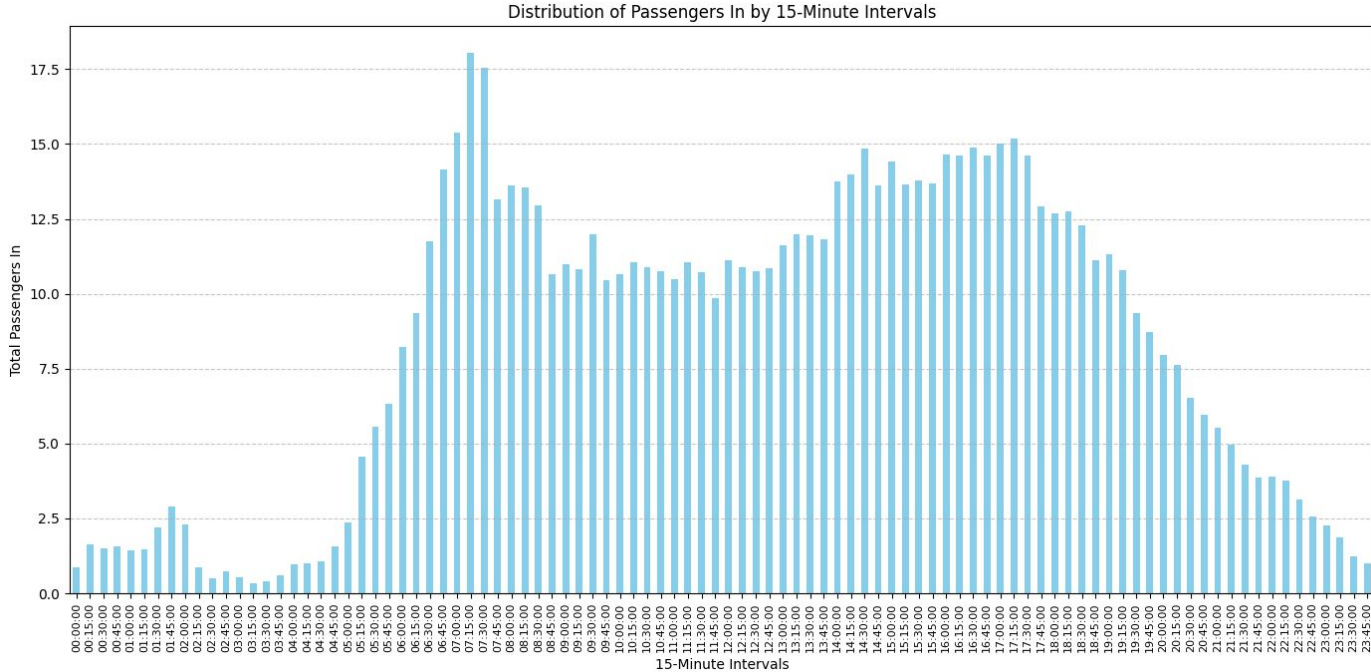
Weighted Percentiles of Travel Time at Every 1 Percentile



Demand Based Pricing Function



Off-peak hours discount



To incorporate the concept of peak and off-peak hours into the travel-time based pricing function, we can introduce a discount mechanism based on urban studies trends, specifically focusing on morning peak hours and off-peak hours.[1]

- Morning Peak Hours: 6:45 AM to 7:45 AM
- Off-Peak Hours: 6:00 AM to 6:40 AM and 7:40 AM to 8:40 AM

Change in earnings

Earnings per year before changes:

$$\sum_{ps} 0.65 \cdot \mathbb{1}_{(0,30)}(t) + 0.9 \cdot \mathbb{1}_{(30,\infty)}(t) = 3801737.8$$

Earnings per year after changes:

$$\sum_{nops} p \cdot (1 - 0.0229) + \sum_{ops} p \cdot 1/2 \cdot 1.0442 = 3801737.8$$

Earnings difference is less than 1% and that does not count passenger amount changes after dynamic pricing

Future plans

- more time travel values (new stops, different time)
- time series analysis of the demand
- get more modern data
- validate the results using using an activity-based demand model SimMobility Pre-day
- provide results to Analytics Department of Vilnius public transport (wait until the end of their vacation)

Literature list

[1] Lovrić, Milan & Raveau, Sebastián & Adnan, Muhammad & Pereira, Francisco & Basak, Kakali & Loganathan, Harish & Ben-Akiva, Moshe. (2016). Evaluating Off-Peak Pricing Strategies in Public Transportation with an Activity-Based Approach. Transportation Research Record Journal of the Transportation Research Board. 2544. 10.3141/2544-02.

Branda, F.; Marozzo, F.; Talia, D. Ticket Sales Prediction and Dynamic Pricing Strategies in Public Transport. Big Data Cogn. Comput. 2020, 4, 36.

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SimMobility Freight: An agent-based urban freight simulator for evaluating logistics solutions,

Transportation Research Part E: Logistics and Transportation Review, Volume 141, 2020, 102017, ISSN 1366-5545

Urbanek, Anna. (2019). Public Transport Fares as an Instrument of Impact on the Travel Behaviour: An Empirical Analysis of the Price Elasticity of Demand. 10.1007/978-3-030-17743-0_9.

Urbanek, Anna & Tomanek, Robert & Dydkowski, Grzegorz. (2018). Taryfy i systemy poboru opłat w miejskim transporcie zbiorowym.